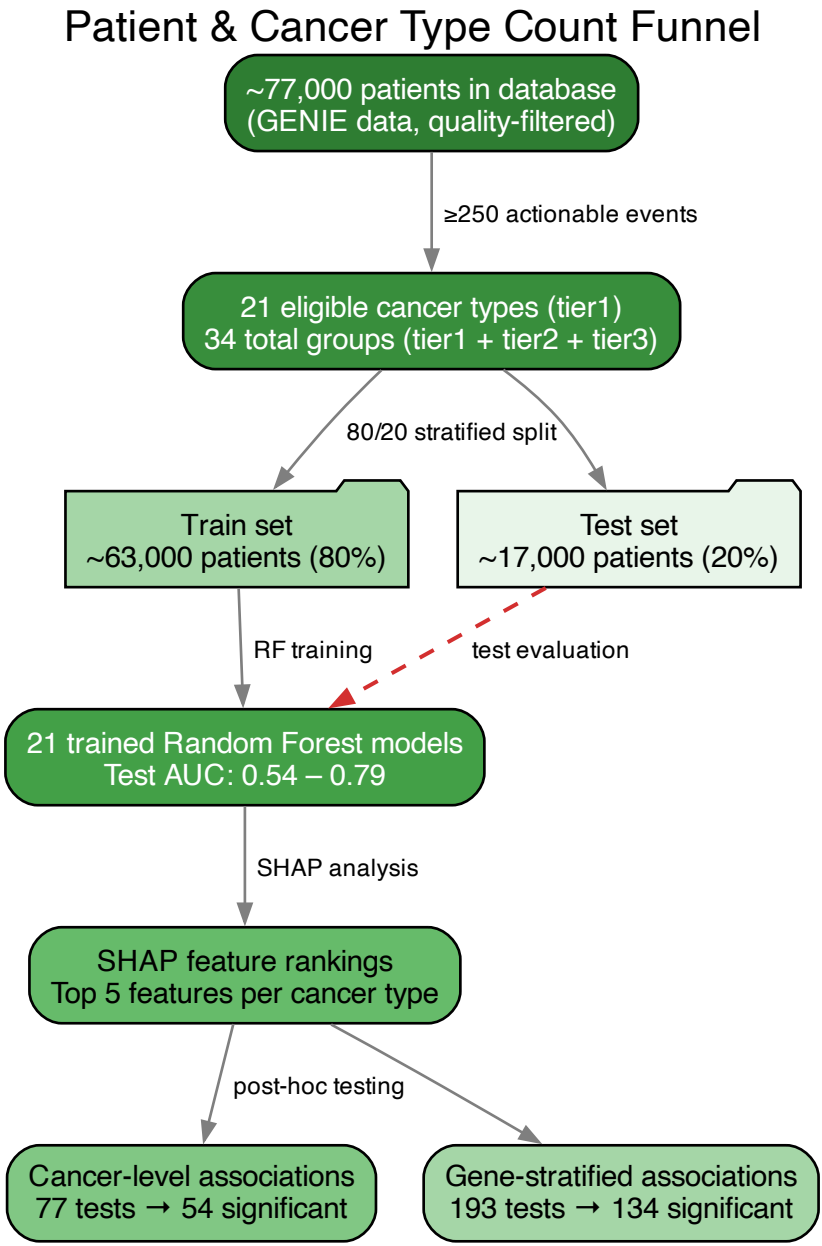
Patient & Cancer Type Count Funnel
~77,000 patients in database
(GENIE data, quality-filtered)
≥250 actionable events
21 eligible cancer types (tier1)
34 total groups (tier1 + tier2 + tier3)
80/20 stratified split
Train set
~63,000 patients (80%)
Test set
~17,000 patients (20%)
RF training
test evaluation
21 trained Random Forest models
Test AUC: 0.54 – 0.79
SHAP analysis
SHAP feature rankings
Top 5 features per cancer type
post-hoc testing
Cancer-level associations
77 tests → 54 significant
Gene-stratified associations
193 tests → 134 significant